

MANDAL, China

WMO: 531490

Lat: 42.5293N

Lon: 110.1166E

Elev: 1223

StdP: 87.47

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-22.0	-32.3	0.2	-14.7	-30.3	0.3	-14.8	14.8	-7.2	13.3	-8.7	3.5	270	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	33.1	15.5	31.5	15.2	29.9	14.7	18.9	26.8	17.9	25.3	17.0	24.5	4.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.8	13.9	21.3	15.6	12.8	20.6	14.3	11.8	20.5	59.3	26.7	55.7	25.5	52.7	24.5	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.2	9.8	-26.8	35.6	2.6	1.8	-28.7	36.9	-30.2	38.0	-31.7	39.0	-33.6	40.3		
(5)			-27.1	20.1	2.6	1.2	-28.9	21.0	-30.4	21.7	-31.9	22.3	-33.7	23.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-12.8	-8.3	-0.6	8.2	14.9	20.7	23.7	21.5	15.4	7.0	-3.0	-10.5		
	DBStd	13.45	4.92	5.58	5.75	5.46	4.72	3.65	2.86	3.18	4.36	5.11	5.65	4.97		
	HDD10.0	2811	706	512	330	97	12	0	0	0	8	119	390	637		
	HDD18.3	4749	964	746	587	305	126	18	1	9	106	353	640	895		
	CDD10.0	1504	0	0	1	42	163	321	424	357	171	26	0	0		
	CDD18.3	402	0	0	0	1	19	89	167	107	18	0	0	0		
	CDH23.3	3921	0	0	0	26	271	896	1630	925	170	4	0	0		
	CDH26.7	1382	0	0	0	2	75	314	655	301	35	0	0	0		
(14) Wind	WSAvg	4.3	4.5	4.2	4.6	4.9	4.9	4.0	3.5	3.3	3.6	4.2	4.7	4.8		
(15) Precipitation	PrecAvg	168	2	2	3	7	11	20	47	44	21	8	3	1		
	PrecMax	326	7	8	11	30	56	62	132	115	69	33	17	5		
	PrecMin	90	0	0	0	0	0	0	10	10	2	0	0	0		
	PrecStd	57	2	2	3	6	13	13	32	26	18	7	4	1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.8	9.1	17.7	26.4	31.5	34.1	36.0	33.7	30.2	23.4	13.6	4.4		
		MCWB	-2.8	1.1	5.4	9.7	12.1	14.7	17.2	15.9	13.4	10.4	4.2	-1.9		
	2%	DB	-0.5	5.2	14.3	23.3	28.8	32.1	33.8	31.7	27.3	20.1	10.3	1.0		
		MCWB	-4.9	-1.4	3.9	8.3	11.4	14.0	16.5	16.0	12.6	8.7	2.4	-3.6		
	5%	DB	-3.1	2.8	11.5	20.6	26.3	30.2	32.0	29.9	25.2	17.7	7.6	-1.4		
		MCWB	-6.6	-2.8	2.1	7.3	10.3	13.7	16.0	15.4	12.0	7.4	1.0	-5.2		
	10%	DB	-5.3	0.5	8.8	17.9	24.0	28.3	30.3	28.2	23.2	15.3	5.4	-3.4		
		MCWB	-8.1	-4.0	0.7	6.0	9.5	13.2	15.6	14.9	11.3	6.2	-0.2	-6.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.5	1.5	6.0	11.3	13.7	17.5	20.3	20.4	16.9	12.1	4.8	-1.7		
		MCDB	2.3	7.3	16.8	21.5	24.4	23.6	28.8	25.9	22.4	20.4	11.7	3.7		
	2%	WB	-4.9	-0.9	3.9	9.3	12.4	16.4	19.1	19.0	14.7	9.6	2.9	-3.5		
		MCDB	-0.8	4.2	13.5	20.6	24.4	24.4	27.6	26.1	21.5	18.0	9.4	0.7		
	5%	WB	-6.6	-2.4	2.4	7.8	11.4	15.5	18.1	17.9	13.6	8.1	1.3	-5.1		
		MCDB	-3.3	2.5	10.6	19.4	23.0	24.7	25.7	25.5	21.7	15.8	7.3	-1.6		
	10%	WB	-8.1	-3.9	1.0	6.5	10.4	14.7	17.4	16.8	12.6	6.8	-0.2	-6.5		
		MCDB	-5.4	0.5	8.5	17.2	21.9	24.5	25.1	23.9	21.0	14.1	5.1	-3.5		
(35) Mean Daily Temperature Range	MDBR		10.4	11.7	12.2	12.2	12.3	11.3	10.7	10.6	11.1	11.2	10.4	9.7		
	5% DB	MCDBR	12.5	13.8	15.2	15.2	15.1	13.7	13.0	12.7	13.3	13.6	12.8	11.8		
		MCWBR	9.3	9.2	8.8	7.3	6.4	4.5	3.8	3.9	5.0	6.5	7.6	8.5		
	5% WB	MCDBR	12.3	13.0	14.3	13.8	12.8	11.0	9.9	9.8	10.8	12.3	12.3	11.7		
		MCWBR	9.3	8.8	8.5	7.0	5.9	4.0	3.6	4.2	4.9	6.5	7.5	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.274	0.309	0.350	0.400	0.436	0.465	0.442	0.426	0.392	0.342	0.296	0.277		
	taud		2.282	2.195	2.099	2.013	1.969	1.976	2.104	2.131	2.151	2.231	2.279	2.253		
	Ebn at Noon		866	883	884	864	840	815	830	830	830	832	826	820		
	Edh at Noon		86	112	140	166	178	178	155	146	133	107	86	81		
(44) All-Sky Solar Radiation	RadAvg		2.39	3.50	4.96	6.16	6.90	6.63	6.44	5.98	5.03	3.92	2.67	2.03		
	RadStd		0.26	0.27	0.32	0.29	0.33	0.38	0.26	0.24	0.24	0.15	0.19	0.26		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.47	-0.74	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	-0.97	N/A	N/A	-0.61	N/A	N/A	N/A	N/A

Nomenclature: See separate page